A Future Approach to Integration in Power Electronics Systems

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***Abstract*—Assemblies of power semiconductor switches and their associated drive circuit are at present available in modules. Upward into the multi-kilowatt range, mixed mode module construction is used. This incorporates monolithic, hybrid, surface mount and wirebond technology. However, a close examination of the applications in motor drives and power supplies indicates that there has been no dramatic volume reduction of the subsystem. The power semiconductor modules have shrunk the power switching part of the converter, but the bulk of the subsystem volume still comprises the associated control, sensing, electromagnetic power passives (inductors, transformers, capacitors) and interconnects. This paper addresses the improvement of power processing technology through advanced integration of power electronics. The goal of a subsystem in a module necessitates this advanced integration, incorporating active switching stages, EMI-filters and electromagnetic power passives into modules. The central philosophy of the technology development research in the National Science Foundation Engineering Research Center for Power Electronic Systems is to advance the state of the art by providing the concept of integrated power electronics modules (IPEMs) for all these functions. The technology underpinning such an IPEM approach is discussed. The fundamental functions in electronic power processing, the materials, processes and integration approaches and future concepts are explained.**

I. Introduction

ELECTRONIC power processing has steadily moved toward achieving the necessary output waveforms by high-frequency synthesis, resulting in important reductions in converter performance, size, weight and cost. At present it appears that in some high frequency power processing technologies, fundamental limits are being reached that will not be overcome without a radical change in the design and implementation of power electronics systems. Over the last decades, the performance of power electronics systems has been driven by improvements in semiconductor components. Moving from bipolar to MOSFET technology [1, 2] has resulted in speed increases that, today, test the limits of structural inductances associated with packaging, as well as thermal handling. Thus, an order of magnitude increase in switching speed, which is possible with new device technologies, will require substantial reductions in structural capacitances and inductances associated with device and system-level packaging.

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For a typical power electronics system, individual power devices are mounted on a heat sink, and the drivers, sensors, and protection circuits are implemented on a printed circuit board and mounted near the power devices. The manufacturing process for such equipment is labor and cost-intensive. In recent years a level of integration has been developed where power semiconductors in die form are mounted on a common substrate and interconnected with wire bonding. In principle, this is a form of discrete miniaturization where the same discrete dies are de-encapsulated and all components placed closer to each other. The number of interconnections and discrete functionality of all components remain as in a converter based on discrete construction. Also, in the recent past, wire bonding technology has seen many improvements in reliability, but the possibilities for three-dimensional integration are still limited, and electromagnetic layout constraints have become dominant. The thermal management in this type of packaging is essentially limited to heat removal from one surface of the die, while the requirements for heat extraction escalate continuously due to the close proximity of the components to each other. This is an indicator of another possible limit in this type of approach.

It is perceived, however, that power electronics modules constitute one of the driving forces towards modularization and integration of power electronics systems. Recent innovations in power modules have been mostly driven by semiconductor development with the help of improved layout and packaging technologies. The development trends for power modules have been focused on increasing current and voltage levels, increasing operational temperature, enhancing reliability and functionality, as well as reducing size, weight and cost [3]. New ideas on power modules include the reduction of the number of interfaces to reduce the number of solder layers and the risk of solder imperfections and thermal fatigue. Another concept that has been around is the elimination of the base plate from the power module, first observed at low power and now migrating to higher power applications. In terms of reliability improvement, devices using low temperature joint (LTJ) technology do not change their thermal resistance after extensive power cycles (>50000), as opposed to devices using low temperature conventional solder techniques that achieve the end of useful life after 30000 power cycles [3].

While semiconductor devices are still one of the dominant barriers for future power system development, devices do not currently pose the fundamental limitation to power conversion

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technology. It has been argued for some time now that it is rather packaging, control, thermal management, and system integration issues that are the dominant technology barriers currently limiting the rapid growth of power conversion applications [4-7]. To address some of these issues, this paper briefly discusses the technology advancements needed to improve the characteristics of power electronics systems, as well as the technologies being developed for integration of multi-kilowatt power electronics. Among the technologies being developed are planar metalization device interconnects, allowing three-dimensional integration of power devices, as well as the integration of power passives to increase the power density, as these dominate the physical size of the system. The technologies being developed will ultimately span a wide range of applications from distributed power systems to motor drives. This paper also discusses results obtained from the various technologies being developed within the research scope of the Center for Power Electronics Systems (CPES), whose mission is to promote an integrated approach to power electronics systems in the form of highly Integrated Power Electronics Modules (IPEMs).

II. A Motivation for High Density Integration

The advantages of incorporating high-density multi-functionality into solids through planar integration technology have been well illustrated by the advances of microelectronics in the past [8]. Although electronic energy processing is different from information processing, the following parallels are important to note:

- Both technologies have electromagnetics as a fundamental limit;
- Both technologies are eventually thermo-mechanically limited (i.e. in terms of interface reliability and loss density);
- Both technologies are materials limited;
- New applications for both are driven by a relentless downward cost spiral.

To evaluate what should be done regarding the integration of power processing, it will be useful to examine the fundamental functions applied in power processing.

A. Functions Performed In Power Processing

Microelectronics integration technology has developed using a single semiconductor material--Silicon. Such exclusive use of Silicon has not yet been successful for power electronics integration. The most important reason for this is the inadequate electromagnetic characteristics of Silicon, thus necessitating a hybrid integration technology for power electronics [9, 10]. Fig. 1 defines the fundamental functions found in any electronic power processor. The switching function is used to control the flow of electromagnetic energy through the processor, the conduction function is used to guide the flow appropriately, while the electromagnetic energy storage function is used to provide energy continuity at the input and output of the processor. The information function takes care of the necessary sensing and relates the spatial and temporal action of the first three functions. Finally, the heat exchange function maintains the thermal integrity of the system. A careful consideration of these fundamental functions to be performed in power processing indicates that only the switching function and the information function can be successfully accomplished in known semiconductors such as Silicon, due to the electromagnetic characteristics required for the conduction and electromagnetic energy storage function.

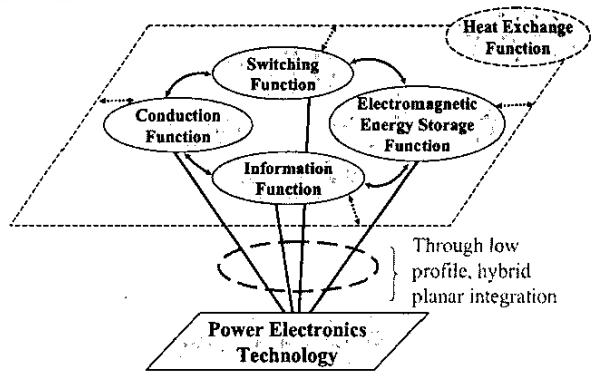


Fig. 1. Fundamental functions in electronic power processing.

A real paradigm shift in integration technology for power electronics can therefore evolve from the development of a technology that uses appropriate planar electromagnetic materials in conjunction with metalization for all interconnects and power semiconductor devices, planar layers for dielectrics and magnetics and their interconnects and stacking of these planar layers in a quasi-3D fashion for high density – such as is not easily achievable with the widely used wire bonding – leading to actual low profile integrated power processing blocks having quasi-planar form factors as modules for power electronic system synthesis [11]. In this low profile construction the power processing will conform to the form factor of the applications it drives (flat screens, flat lamps, flat laptops, linear propulsion systems, etc.), and at the same time it will increase the surface-to-volume ratio very favorably for thermal management.

B. Technologies and Process Level Integration

Furthermore, present manufacturing processes of the different types of discrete components and their packaging as used in power electronics are not compatible with integration into one production line. Examples of these technologies are the power semiconductor processing, coil and transformer winding, capacitor winding and other capacitor manufacturing methods, resistor manufacturing technologies, and the construction of power electronic converters in discrete component technology by wiring harness methods.

Compatible processes that can be incorporated into one integrated production line are needed. As the nature of the integrated module will be hybrid, however, it has to be assumed that there will be a materials based manufacturing process (for producing power semiconductor dies, magnetic sheets, dielectric sheets, substrate sheets, for example)

preceding the planar metalization, the large area electrical and thermal contacting, the integrated planar power passive formation and the layer stacking and packaging operations [12]. We have developed our technologies with this approach in mind. To achieve this we have placed a limitation on the nature of the processes we use. Only the following processes are implemented in the technologies discussed further on, since they are all compatible with already existing manufacturing process for integrated modules:

- Planar metalization technology by sputtering and electroplating;
- Reflow soldering for high quality large area electrical-thermal interconnects;
- Low temperature sintering for high quality large area electrical-thermal interconnects;
- Photolithography and wet chemistry for pattern definition and etching;
- Plasma cleaning as intermediate process steps for interface improvement;
- Laser machining and trimming as mechanical formation steps;
- Encapsulation technology for electrical, mechanical and thermal integrity.

The use of selections of these process steps will be found in each of the technologies described further in this paper.

As shown in Fig. 2, consideration should further be given to integration at the process technology level when developing the processes. Fig. 2 characterizes wirebonding as an essentially sequential interconnect/integration technology when applied to power electronic modules. As also shown in Fig. 2, it is desirable to achieve simultaneous execution of a large number of parallel steps when considering interconnection. Examples of this are Embedded Power as a power switching stage integration technology and the integration of Electromagnetic Power Passives as will be discussed in Section III of this paper.

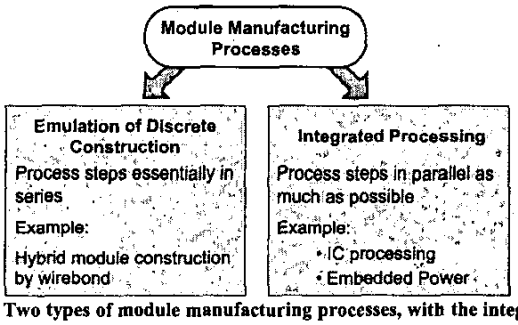


Fig. 2. Two types of module manufacturing processes, with the integrated processing showing process level integration.

C. *Partitioning for Integration*

Although the processes used in bybrid integration of electronic power processing have now been chosen to be compatible with an integrated production, it appears that other considerations dictate partitioning the integration of the system into different modules. As shown in Fig. 3, thermal considerations favor the partitioning into Active IPEMs for the power switching stages and Passive IPEMs for the electromagnetic energy storage and conversion. While the loss energy density in the switching devices is high, the loss energy density in the passive devices is low. This requires different thermal approaches for optimum module performance.

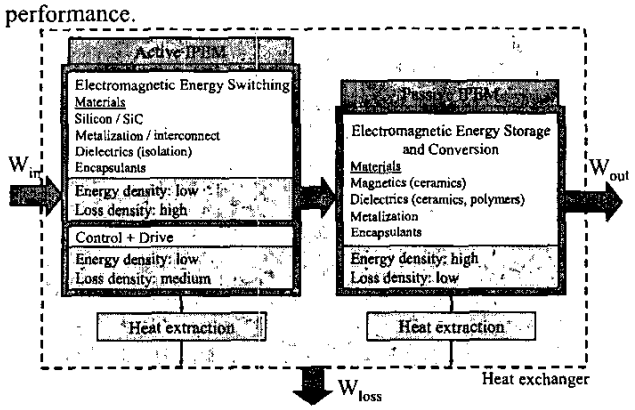


Fig. 3. Power electronics converter partitioning into active and passive IPEMs for optimum performance of integrated systems.

Furthermore, as will be discussed in section III, the electromagnetic characteristics of the passive IPEMs are not suited to the integration of EMI filters when constructed according to the requirements for optimum module performance for storing and processing electromagnetic energy. Consequently, a third type of IPEM for EMI filtering becomes necessary.

III. High Density Low Profile Integrated Power Electronics Structures

As examples of achieving high-density power processing structures with potential for incorporation into 3-D integration in quasi-planar modules, the following approaches were developed. These approaches use some or all of the above steps to produce integrated active power switching modules and integrated electromagnetic power passive modules for building integrated power electronic converters for currents up to 50A and voltages up to 800V.

A. *Integrated switching modules in Embedded Power*

"Embedded Power (EP)" refers to an integration technology, which was developed in-house for packaging power electronics switching (active) modules, incorporating multiple bare power chips, such as IGBTs, MOSFETs and diodes, and associated components for drive, control and protection [13]. The power chips are finally embedded in a three-dimensional (3-D), multilayer interconnect structure by sandwiching the power bus network and integrated circuitry. In addition, the planar metallization interconnection and ceramic chip carrier lead to functional integration, while planar integration processes such as photolithography, thin- and thick-films deposition, allow cost-effective automation during manufacture.

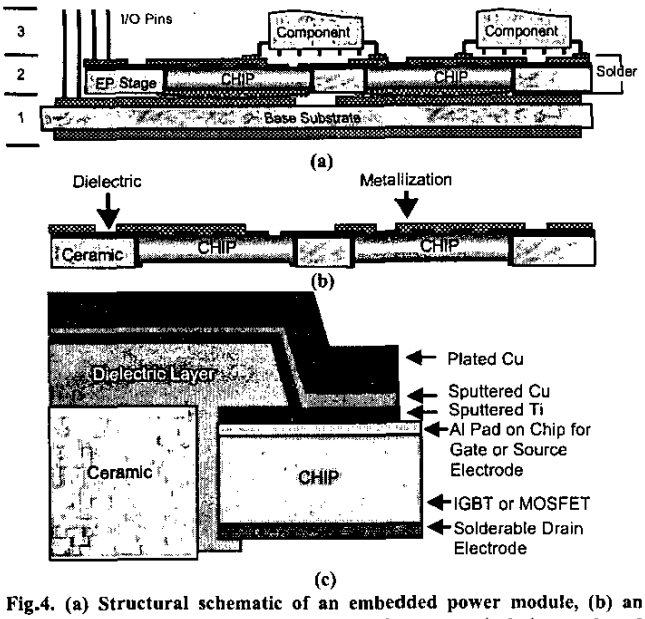


Fig.4. (a) Structural schematic of an embedded power module, (b) an integrated power chips stage, and (c) chip mount, isolation and pad contact.

Used in the assembly of power electronics modules, most power semiconductor chips have an inherent vertical structure in which the metallization input/output (I/O) electrodes (pads) are arranged on two sides of the chips. Usually, the gate, source (or emitter) pads are located on the top surface with the thin film metal aluminum (Al) terminals and are ready for ultrasonic bonding. The drain (or collector) electrode is the deposited metallization (mostly Ag or Au) on the bottom of the chip, which is ready to be soldered to a base substrate. This vertical structure enables the building of a sandwich-type 3-D multi-chip module (MCM) construction. Fig. 4 (a) illustrates the cross-section view of a power electronics module designed in this way. It consists of three layers: (1) base substrate, (2) EP stage and (3) components, connected by solder layers. In the structure, the EP stage - an integrated power chips component - is built on a flat ceramic frame, as shown in Fig. 4 (b), with the chips embedded. The dielectric is used as an interlayer to protect the chip with defined vias on the top pads of the chips. The metallization is for the interconnection of multiple power chips through metallurgical contact to aluminum pads on the chips through vias in the dielectric layer; this replaces the bond wires in the conventional module, and is also used as the next-level interconnection for the attachment of associated components (layer 3 in Fig. 4 (a)). Layer 1, the base substrate, can be a direct-bond copper (DBC) ceramic substrate for the bottom solder interconnects of multiple chips and also offers a thermal downward path. After stacking three layers by soldering, the structure offers several advantages, such as sandwiched bus layout, closer component attachment, shorter interconnects, and a solid block. This arrangement leads to higher power density, improved electrical performance and effective thermal dissipation. Fig. 4 (c) shows the details of the pad contact, mounting and isolation. The original bare chips encompass the Al pads, the solderable bottom electrode and passivation layer outside the pad surfaces. An adhesive dielectric is used to affix the chips through filling the gaps between the chip edges and the openings in the ceramic frame. The bond layer is comprised of Ti and Cu thin-film layers, which provide good adhesion and low film stress, as well as excellent electrical/thermal conduction. For carrying high current, a thicker (>5mils) Cu layer is designed over the thin-film Cu layer.

The whole packaging process is designed as a two-step manufacturing operation: the integration of multiple chips and the 3-D assembly.

Fig.5 illustrates the process flow of the embedded power chips stage. Step 1 involves the preparation of the ceramic frame (chips carrier), which is achieved by laser cutting a ceramic plate. The openings formed are ready for the mounting of the power chips. For accurate alignment of the following processes, multiple tiny tips are formed around the edges of each opening to fix the chips tightly and to provide 500μm-wide gaps. Step 2, mounting the chips, involves filling these gaps completely with a dielectric paste, then using temperature curing (150°C/7mins). Now a flat substrate with co-planar chips is formed, which allows the subsequent planar processing. The whole surface is coated with a dielectric interlayer, as shown in step 3, which defines the vias on the pads of the chips by using a screen-printing method with a prepared screen pattern. The screened dielectric paste also needs a curing of 200°C/30mins to form a flat, good adhesive layer. Then, in step 4, the metallization multilayers are deposited on the whole surface which forms contacts with the Al pads on the chips through via holes in the interlayer. Conventional sputtering is employed to deposit thin-film Ti and Cu layers. Electroplating is used to form a thicker Cu deposition. Finally, in step 5, the metallization layer is patterned to form both the power and the control circuits as well as their I/O terminals.

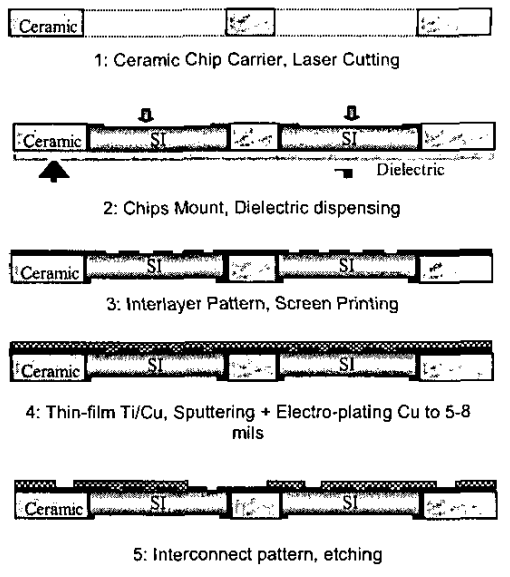


Fig. 5. The process steps of the embedded power chips stage.

B. *Integrated Electromagnetic Power Passive Modules*

In order to integrate the electromagnetic power passive components used in power electronic converters into modules, additional technology had to be innovated. This integration technology for power passives can best be described by first considering a simple bifilar spiral winding as shown in Fig. 6(a). This structure consists of two windings (A-C and B-D), separated by a dielectric material. This resultant structure has distributed inductance and capacitance and is an electromagnetically integrated LC-resonant structure for which equivalent circuit characteristics depend on the external connections (Fig. 6(a)). Even more complex integrated structures can be realized by adding more winding layers. This has been demonstrated with an integrated resonant transformer structure (L-L-C-T) as shown in Fig. 6(b) and (c) [14]. Design of these structures requires deliberate increase and modification of naturally existing structural impedances to realize a particular equivalent circuit function – for example the increase of the intra-winding capacitance to form the LC resonant structure. The classical term "parasitics" therefore no longer applies and all the higher-order impedances are rather referred to as structural impedances. The spiral integrated power passive structure has been developed over a number of years with advances in electromagnetic modeling, design and loss determination [4, 5, 15]. Loss modeling now includes core losses, dielectric losses, and skin and proximity effects, and conduction losses for non-sinusoidal voltages and currents. The improved models have led to integrated structures with power densities in excess of 30W/cm^3 operated at frequencies of up to 1MHz [16].

This developed technology for integrated passive components has mostly been implemented in resonant converter applications [14]. However, its application in a non-resonant PWM converter has also been demonstrated [17-19], as will be outlined in the subsequent systems discussion in IV. In this application, the planar passive integration technology, together with the planar integrated magnetics technology were combined to integrate all of the high frequency passive components in a 1 kW Asymmetrical Half-Bridge DC/DC Converter (AHBC) for a Distributed Power System (DPS) application. The circuit diagram of the AHBC is shown in Fig. 12. The current-doubler inductors and the isolation transformer are not magnetically coupled, but can be integrated into two separate structures by splitting the isolation transformer and utilizing the equivalent magnetizing inductances to realize the current doubler inductors (reflected to the secondary). These two magnetic structures can in turn be integrated into one physical structure through integrated magnetics technology [17]. A cross-section, reluctance diagram and exploded view of this resultant first generation spiral integrated passive module are shown in Fig. 7. The I-core shares the flux for both of the integrated modules $LLCT_1$ and $LLCT_2$ and the AC flux is partially cancelled in the shared I-core as shown in Fig. 7(b). The relatively large dc decoupling capacitor is integrated into both the primary windings of the structure, using very high permittivity (Er > 12000) ceramics (Fig. 7(c)).

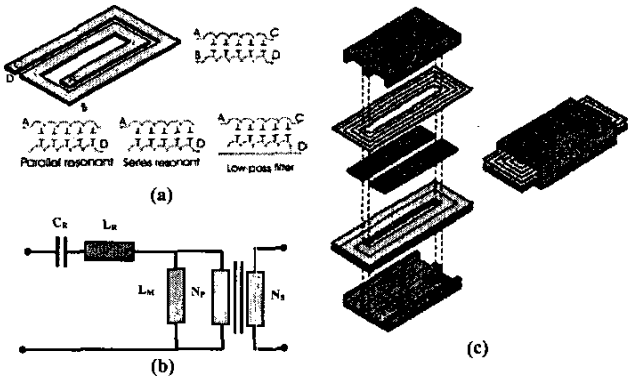


Fig. 6. (a) Spiral integrated LC structure with distributed capacitance and possible external connection configurations, (b) simplified equivalent circuit, and (c) exploded view.

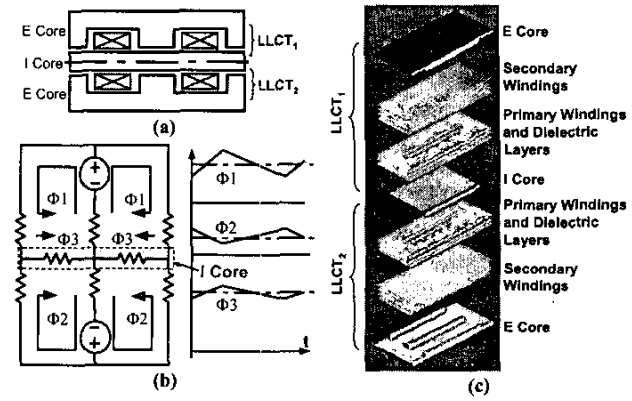


Fig. 7. (a) Cross section, (b) reluctance diagram, and (c) exploded view of non-resonant spiral integrated passive structure with integrated magnetics.

C. *Integrated EMI Filter IPEMs*

Integration of power electromagnetic components, such as high frequency transformers, resonant/choke inductors and resonant/blocking capacitors, for Switch Mode Power Supplies (SMPS) has been studied in order to reduce the count, volume and profile of the components to increase the power density of the converter. Its applications in resonant and non-resonant SMPS have been reported [14-19]. However, very little work has been done on the subject of filter integration. To bridge this gap, the integration of an EMI filter for a DPS front-end converter was studied [20]. By applying the low-pass filter configuration of the planar Integrated L-C structure, as illustrated in Fig. 6, an EMI filter shown in Fig. 8 can be integrated into a single planar module.

Since functions of the high frequency power passive components and the EMI filter are different in SMPS circuits, their requirements are accordingly different. For high frequency power passive components, such as resonant inductors, capacitors and transformers, the major constraint is thermal. Although the integration can reduce the component size and increase the power density, the loss density increases at the same time. To alleviate the thermal problem, the

integrated module normally needs to be designed to minimize the losses, especially for switching frequency and the frequencies above. In addition, since the most interesting frequency range of these modules is close to their self-resonant frequency, the high frequency characteristics of these modules are normally not the major concern. However, for EMI filters, the requirements are quite different. Firstly, since the function of the EMI filter is to attenuate the noise at switching frequency and its harmonic frequency, the fundamental self-resonant frequency of the integrated EMI filter, which is the corner frequency of the low pass filter, is normally much lower than the switching frequency. The high frequency characteristics now become the major concern for the filter design and construction. Secondly, from the point of view of losses, high frequency losses are desirable to damp the high frequency noise. Since the power current passing through the filter is at line frequency, increasing high frequency losses will not add much to the thermal burden as long as the losses at line frequency are kept small. From the above analysis, it could be concluded that the major technology requirements for the integrated EMI filter include improving high frequency performance, which could be implemented by reducing the equivalent parallel capacitance (EPC) of the inductors and the equivalent series inductance (ESL) of the equivalent capacitors of the integrated module, and increasing high frequency losses and maintaining low losses at line frequency.

To reduce the EPC of the integrated inductors, a narrow and thick conductor shape is more suitable than the conventional wide and thin conductor shape, which is normally preferable in other applications due to the reduced high frequency eddy current losses. For EMI filters, the narrow and thick conductor shape can reduce the conductor surface area, hence the structural winding capacitance can be reduced. It can also increase the high frequency eddy current winding losses to increase the high frequency damping factor.

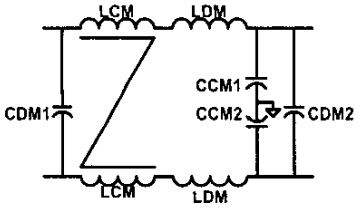


Fig. 8. Schematic of a typical EMI Filter.

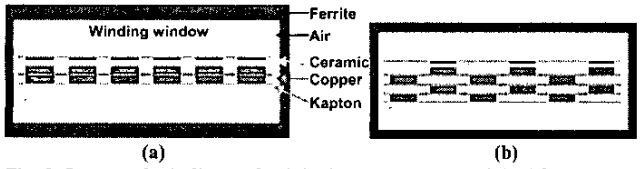


Fig. 9. Staggered winding and original structures: (a) original Structure, and (b) staggered winding structure.

To further reduce EPC, a staggered winding structure is used, as shown in Fig. 9. It greatly increases the distance between layers, hence the structural winding capacitance can be greatly reduced. By applying these technologies, the EPC of a constructed prototype is reduced from originally around 100pF to less than 10pF.

The ESL of the integrated capacitors can be reduced by using the transmission line connection method, since it shifts the ESL out of the capacitor branch. To further increase the high frequency eddy current loss, multi-metal metallization technology is introduced. Instead of using pure copper as the conductor material, a Nickel-Copper-Nickel combination is used. Nickel is not only a higher resistivity material than copper, but its increased permeability gives a much smaller skin depth at the same frequency. Hence the high frequency eddy current loss can be greatly increased. The low frequency power current will automatically flow through the bulk copper layer, which will give low conduction loss and low temperature rise.

IV. Integrated Power Electronics Systems – the DPS Example

A. DPS Implementation Using IPEMs: System Level View

The use of distributed power systems (DPS) for computer and telecommunication applications has opened up the opportunity to develop a standardized modular approach to power processing, which will improve the design and manufacturing processes significantly, as well as enhance the electrical system performance. For this reason, a DPS has been chosen to demonstrate the advantages of integrating power electronics systems, using a selection of the technologies described previously.

The structure of a typical DPS is shown in Fig. 10, while a building block of a front-end converter is illustrated in details in Fig. 12. A typical three-dimensional solid-body model representation for this type of system is shown in Fig. 11(a). As observed, the existing approach requires discrete components, which makes it difficult to optimize space usage and to increase the system power density. The solid-body model shows that the components of the front-end converter are distributed on the printed circuit board, while Cu traces are routed to distribute power and control signals to all devices. This layout approach presents severe limitations from the electrical and thermal viewpoints. Fig. 11(b) shows the top view of the 3-D solid-body model. The major components are identified in the same figure, as well as the critical paths defining which components must be placed next to each other in order to reduce structural inductances. The inductance formed by any of the critical loops will generate voltage overshoot across the PFC and DC/DC switches during turn-off.

Although bringing devices together helps reduce the structural inductances, a clear limitation to this packaging approach lies in the space optimization, which is limited by the form factor of the various components. Therefore, the structural inductance of the packaging approach shown in Fig. 11(a) becomes a limiting factor to increasing the power level, switching frequency and consequently power density. The

only way to overcome this problem is to minimize the inductance of the critical paths by seeking integrated approaches for power devices and components. The realization of an integrated power electronics system requires advances in technologies, which depend upon finding solutions to deal with the multi-disciplinary issues in materials, electromagnetic compatibility and thermal management.

The end objective is to integrate the functions and components for both the PFC and DC/DC converters shown in Fig. 10. In correspondence with the discussion on partitioning in II, Fig. 12 identifies the three blocks as an EMI filter IPEM, an active integrated power electronics module (active IPEM) and as a passive integrated power electronics module (passive IPEM). The active IPEM represents the integration of power MOSFETs and gate drivers, for the PFC and DC/DC converter part. The main goals of the overall integration are to reduce parts count, increase power density, develop a modular approach, improve thermal management, and reduce the overall number of interconnections at the system level.

The starting point to designing an integrated system is to define specifications at the system and module levels. For this reason, Table I shows the specifications for the DC/DC converter (system level) operated at 200kHz, while Table II and Table III describe the requirements for the active and passive IPEMs (module level).

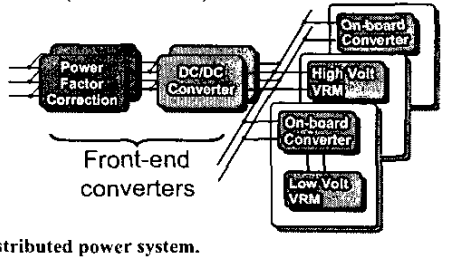
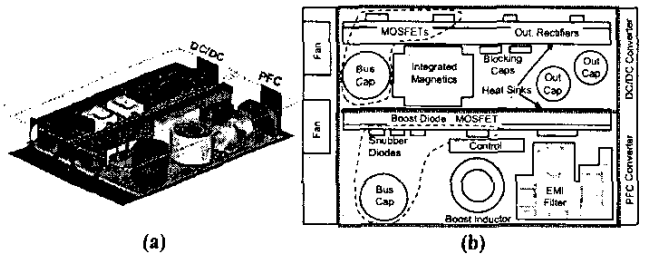


Fig. 10. Distributed power system.



(a) (b)

Fig. 11. Typical component layout for discrete DPS applications: (a) 3-D solid-body model and (b) top view and critical paths of a typical discrete DPS front-end converter.

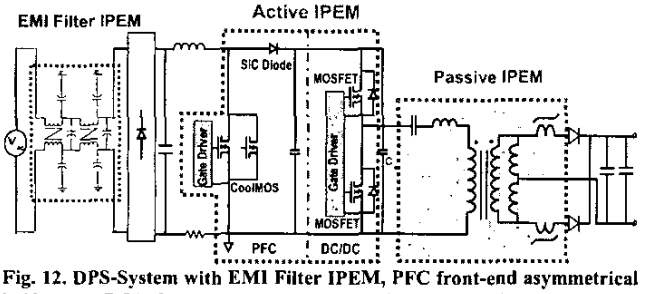


Fig. 12. DPS-System with EMI Filter IPEM, PFC front-end asymmetrical half-bridge DC/DC converter (AHBC) as active IPEM and passive IPEM.

Table I. DC/DC converter specifications.

Parameter	Specification
Input voltage	300 V – 415 V
Output voltage	48 V ± 10%
Output voltage ripple (pk-pk)	480 mV
Isolation (output to ground)	>10kΩ
Output power	1 kW
Ambient temperature	50°C
Air flow at 5000ft	20 CFM at 50°C amb. 15 CFM at 35°C amb
Safety	UL 60950
EMI system-level	EN55022 Class B

Table II. Active IPEM requirements.

Parameter	Specification
DC bus voltage (500 device)	400V
Surge of DC Bus voltage	415V
Power terminal current	25A
Junction temperature (max.)	150°C
Maximum operating junction temperature	125°C
Operating case temperature	-20°C -100°C
Isolation voltage (case-to-terminal at 60Hz)	>2500V
Power terminal leakage current	<500μA
Maximum switching frequency	500kHz

Table III. Passive IPEM requirements.

Parameter	Specification
DC blocking capacitance	>2 μF
Leakage inductance (ZVS range 50%-100%)	2 μH
Magnetizing inductance	45 μH
Safety	UL 60950

In the past, several approaches had already been developed by others for integrated packaging of power modules. Previous work reported a MCM-D package for power applications [21], where a Silicon chip with one power transistor was used as a MCM-D substrate onto which the gate driver was mounted using flip-chip technology. A high level of integration was accomplished, but the power bus was still interconnected to the base substrate using bond wires. This also occurred in the IPM packaging technology, where power chips were mounted and interconnected with bond wires onto a high-density substrate [22]. The concepts of embedded chips and bump-less bonding were also previously investigated for microelectronics packaging applications [23]. The planar device metalization technology that was developed in CPES and described previously namely Embedded Power [13], is a 3-D multi-layer integrated packaging technology that sandwiches power bus structure and integrated circuitry, and its implementation will be described subsequently.

B. Active IPEMs in the DPS

In the DPS front-end converter described previously, two power-switching stages were employed respectively in the FPC and DC/DC conversion parts [24]. These two switching stages are cascaded serially in the converter, as shown in Fig. 12, where the two stages comprise the active IPEM part. The stages operate at bus voltage of 400 V_{dc} and power rating of 1kW.

The PFC stage is designed for a single switch continuous-current-mode (CCM) converter. An advanced CoolMOSFET and SiC diode set, in a boost configuration, is chosen for switching at 400 kHz. This frequency ensures a good tradeoff between the size of the electromagnetic interference (EMI)

filter, the boost inductor and the converter efficiency. For good thermal management, two parallel CoolMOSFETs and two parallel SiC diodes are adopted. A high-frequency bus capacitor is designed to decouple the possible parasitics of the external power connector. The maximum operation junction temperature is limited to less than 125°C.

The half-bridge power switching stage is designed for a 400V/48V DC/DC converter. It is composed of two MOSFETs in a totem-pole configuration. The maximum operating junction temperature is limited to less than 125°C. A high frequency capacitor is employed to decouple the parasitics of the connectors.

For high-density integration, the gate drive will be assembled on a hybrid circuit with an Al_2O_3 substrate made by thick-film technology in this packaging version. The bare driver chips and components are mounted onto it through wirebond and surface mounting. This subassembly will be considered as a single component in packaging.

1) Integrated Power Chip Stages

Fig. 13 presents the process series of the PFC stage manufacture. Four openings have been formed in a 625μm-thick Al_2O_3 ceramic plate by laser machining (Fig. 13(a)). Two large openings, each with eight tiny tips around, are for the mounting of MOSFETs, and the two small ones are for diodes. The gap width between the edges of the chip and the edges of the opening is 0.5mm. It is filled by the dielectric paste to affix the chips in the openings (Fig. 13(b)). The curing of the dielectric is performed in an oven for seven minutes at 150°C. The tips ensure that the chips are located in the accurate positions. After screen-printing and curing (200°C/30mins) of the dielectric paste, six via holes are formed on the source pad, and one on the gate pad of each MOSFET chip. The source hole is a 1mm diameter circle, and the gate hole is a 0.3mm square. A 1mm-diameter circular hole is also formed on top of each diode chip. Then, the metallization layer is deposited on the surface and etched with the designed pattern (Fig. 13(c)), which forms a large CoolMOS source and diode anode on the top of the ceramic frame, as well as the wiring circuit for the attachment of the gate drive circuit, bus capacitor, and input pins. While on the backside, the CoolMOS drains and diode cathodes of the chips are directly exposed for soldering interconnection to the substrate (Fig. 13(d)). The ceramic frame-based stage incorporates all PFC power chips and the planar interconnect circuit. Its size is $28.45\times23.12mm^2$.

Performing the same process steps, the two MOSFET chips are integrated in a same Al_2O_3 ceramic frame, as shown in Fig. 14. Its size measures $28.45\times27.32mm^2$. The top metallization pattern (Fig. 14(a)) includes large source electrodes and small gate single input electrodes, as well as the bus capacitor trace and the input signal pins' traces. The signal and power terminals around the MOSFET are physically separated. The drain electrodes on the chips' bottoms are exposed. As the discrete components, the stages are ready for the next level assembly.

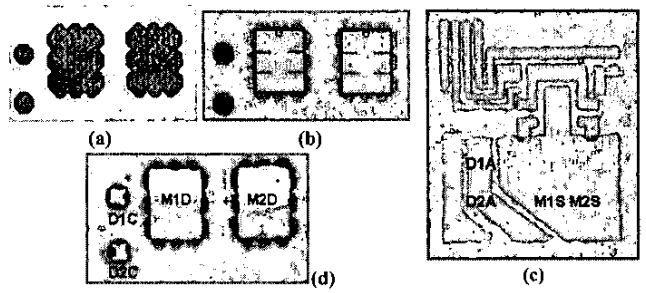


Fig. 13. Fabrication process for the integrated multi-chip embedded power stage: (a) ceramic frame, (b) chip mount, (c) metallization pattern on the topside, and (d) backside view of the stage.

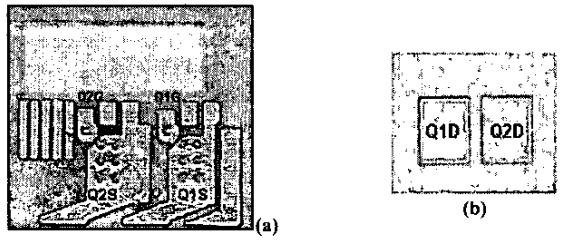


Fig.14. Fabrication for the DC/DC integrated multichip stage: (a) topside metallization, and (b) backside-chip bottom metallization.

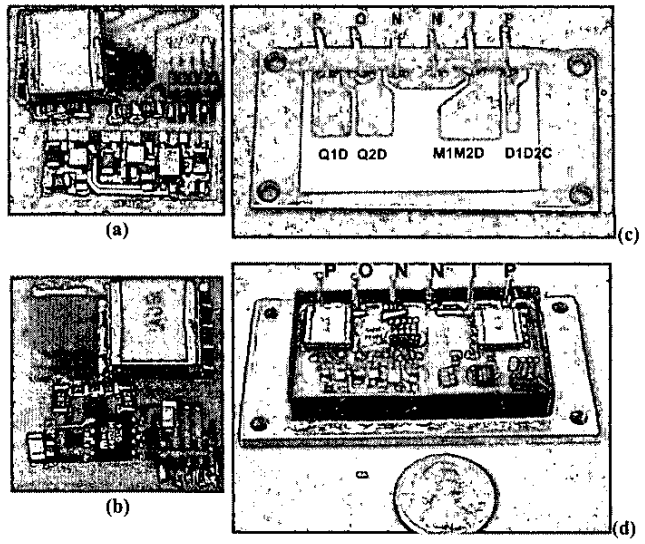


Fig. 15. IPEM assembly from the integrated chip stages: (a) associated components mounted on the top metallization of DC/DC stage, (b) associated components mounted on the top metallization of PFC stage, (c) patterned DBC base substrate attached on a Cu heat spreader plate, and (d) a full PFC+DC/DC IPEM mounted on the base substrate and encapsulated.

2) Assembly of a Subsystem IPEM

The built power chips stages can be further connected on both surfaces by soldering. The associated components can be mounted onto the top metallization. Fig. 15(a) shows that the bus capacitor, components for gate drive, and input pins are mounted onto the top metallization of the PFC stage. Fig. 15(b) shows that the gate-drive hybrid circuit board, bus capacitor, and input signal pins are mounted onto the top metallization pattern of the DC/DC stage. For the backside stacking interconnection, an Al_2O_3 DBC substrate of $28.45\times50.44mm^2$ is patterned on the topside with compact

traces for each chip electrode, as shown in Fig. 13(c). The bottom side is wholly soldered onto a heat spreader made of Cu plate. The Cu posts are mounted on the top traces and form the power terminals.

When the DC/DC and PFC boards shown in Figs. 15(a) and (b) are soldered on the top surface of the DBC substrate, a stacked MCM has been constructed. Fig. 15(d) shows such a 3-D IPEM module encapsulated in a plastic case with silicon gel. The power terminals finally connect the metallization traces on top of the DBC substrate and on top of the chip stages, which are connected with the electrodes of the chips. The module shows the top pins as electrical input and output terminals and the bottom Cu plate as the thermal management interface.

3) Electrical Characterization

The electrical performance of the module was experimentally evaluated. The static characteristics of the power devices in the modules included insulation capability and contact resistance of the metallization interconnection. The switching performance was detailed because of its strong dependence on the parasitics of the packaging structure. The measurements of PFC and DC/DC IPEMs were both carried out. For the PFC part, a testbed was set up in the configuration of a complete boost converter.

Fig. 16 illustrates the efficiency comparison of the PFC IPEM and a similar discrete PFC. At an input voltage ranging from 150 to 250Vac, the IPEM version increases the converter efficiency by 0.5-1.2 percentage points, illustrating the improved switching and conduction performance.

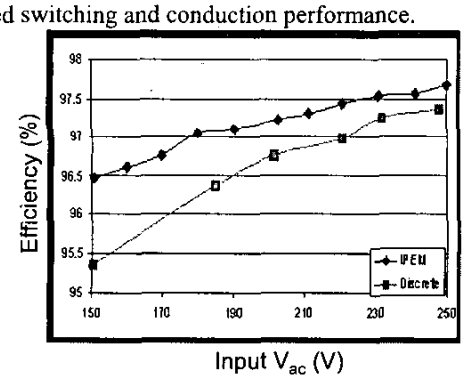


Fig. 16. The efficiency comparison of the PFC converters.

The measurements of the DC/DC IPEM were carried out, using a setup with LC loads, at 400V of DC-biased and 12A peak current (in inductance), with f_s=200kHz, t_r=23.6ns and t_f=24.4ns. For comparison, the wirebond and the EP modules were measured under the same conditions.

The Maxwell Q3D parameter extractor was employed, which is based on the partial element equivalent circuit (PEEC) method, using Maxwell's integral equations instead of differential equations. The details of this method have been described in other work [25]. Table IV lists the DC/DC IPEM's main parasitic parameters and the comparison to the wirebond module. L_D and L_S represent the structural inductance corresponding to the drain and source interconnections, respectively. They are connected in series in the power path and greatly affect the power switching process. The structural inductance of the IPEM has 25% of the value compared to the wire-bond interconnect. The parasitic capacitors between the terminals (Pins P, O and N in Fig. 15) and the ground will affect the system's EMI performance. In addition, the turn-off loss of the IPEM shows reduction from 48μJ to 25μJ. The footprint is reduced 47%. The thermal dissipation capability is improved from 5W/cm^3 to 9W/cm^3.

Table IV. Main parameters in DC/DC IPEM compared to a discrete version

Parameters	Discrete	IPEM
L_D	17.5nH	4.4nH
L_S	16.9nH	4.3nH
$C_{p\text{-ground}}$	25pF	20pF
$C_{o\text{-ground}}$	23pF	20pF
$C_{n\text{-ground}}$	16pF	5.1pF
Loss (E_{off})	48μJ	25μJ
Footprint	40×30mm^2	26×28mm^2
Thermal Capability	5W/cm^3	9W/cm^3

C. Design and Implementation of Passive IPEMs in the DPS

For the design of the second generation passive IPEM, constraints and parameters for all components were obtained from the system requirements and circuit analyses, as illustrated in Table IV. Because of the current doubler configuration, the structure of the passive IPEM was now realized by stacking two transformers and using only one DC blocking capacitor, as illustrated in Fig. 17(a). The transformers are built with two planar E-cores that share a common I-core, as detailed in Fig. 17(b). The DC blocking capacitor of the AHBC is now implemented in only transformer T_1 using the hybrid winding technology [17, 26]. This technology is implemented using Cu traces on both sides of the winding and a dielectric layer placed in the middle to enhance the capacitive component of that winding. The transformer T_2 is a conventional planar low-profile transformer. As already mentioned, the inductances of the current doubler output filter are realized by the magnetizing inductances of both transformers. Fig. 17(c) shows a picture of the final generation 2 passive IPEM implemented for the AHBC.

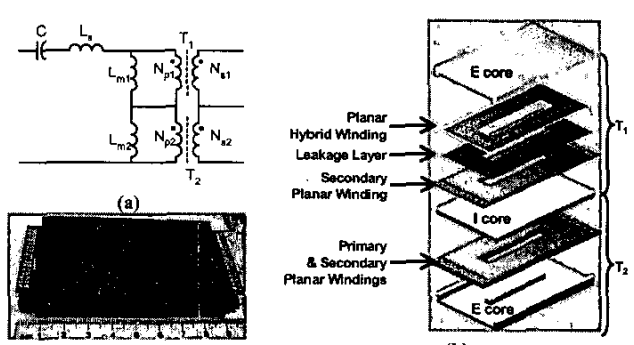


Fig. 17. Components of the passive IPEM: (a) equivalent circuit, (b) exploded view of the passive IPEM, and (c) 1kW prototype.

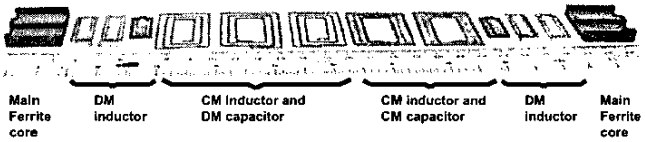
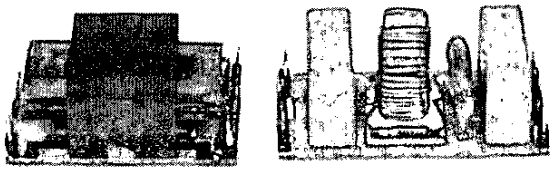


Fig. 18. Exploded view for the constructed integrated EMI filter prototype.

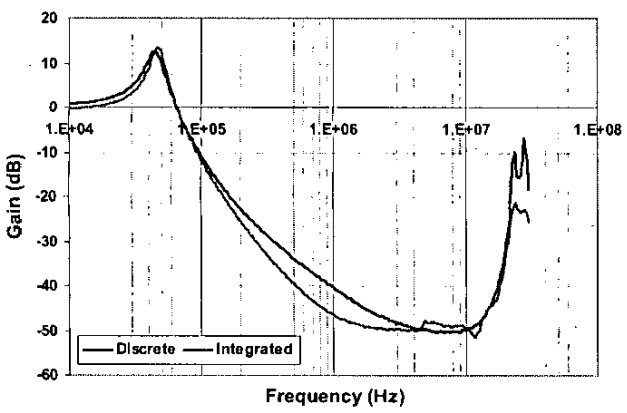


(a) (b)

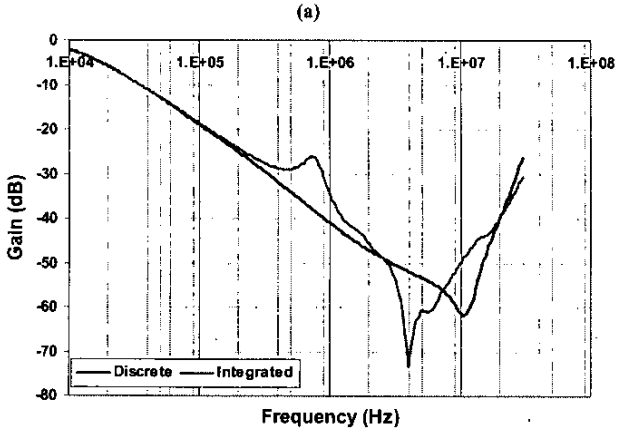
Fig. 19 EMI filter comparison: (a) integrated one and (b) discrete one.

Table V. Parameter comparison of integrated and discrete EMI filters

Parameters	Integrated	Discrete
L_{CM} (mH)	*3.1*	*3.3*
C_{CM} (nF)	*6*	*6.6*
L_{DM} (μH)	*20*	*16.7*
C_{DM} (μF)	*0.6*	*0.56*
EPC (pF)	*3*	*12*
No. of components	*1*	*5*
Profile (cm)	*1.6*	*2.6*
Volume (cm^3)	*27.4*	*39.8*



(a)



(b)

Fig. 20. Measurement comparison of the discrete and integrated EMI filters: (a) CM transfer gains, and (b) DM transfer gains.

D. EMI Filter IPEM for DPS

An EMI filter prototype for DPS was designed and constructed by using the above-mentioned technologies, as shown in Figs. 18-19 [26]. To evaluate its performance, a discrete EMI filter with the same component values is also constructed. The parameters comparison is shown in Table V. Their CM and DM small signal transfer gains were measured by using a HP 4194A Impedance/Gain-phase Analyzer. The measured characteristics are shown in Fig. 20. From these measurement results, it can be concluded that the integrated EMI filter has the same function as the discrete, but much better high frequency CM characteristics, similar high frequency DM characteristics, smaller size, volume and profile.

E. Integrated DPS Converter

Using the proceeding discussed subsystem IPEMs, a complete DPS front-end converter was constructed. Fig. 21 shows the developed prototype. The efficiency of the converter is improved by more than 2% for the wide input AC voltage of from 90V to 270V, compared to the converter constructed with discrete components. Fig. 22 shows these experimental results.

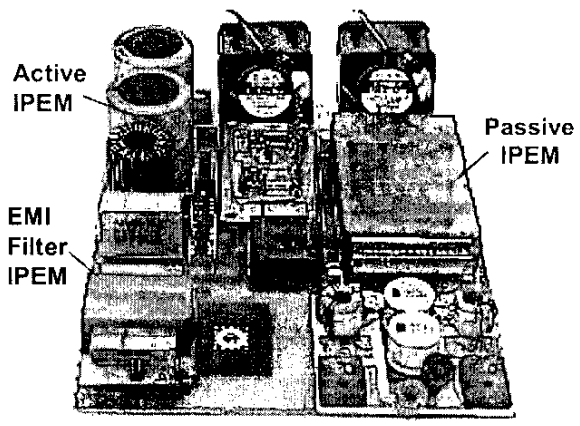


Fig. 21. The construction of the integrated DPS converter.

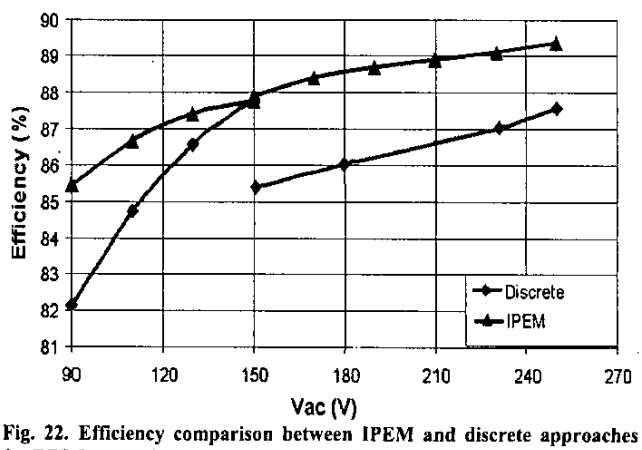


Fig. 22. Efficiency comparison between IPEM and discrete approaches for DPS front-end converters.

V. INTEGRATED DESIGN METHODOLOGY

A multidisciplinary approach to optimizing materials utilization, packaging and thermal management is necessary to realize an integrated system. The entire procedure for designing an IPEM, including layout, fabrication and systems applications necessitates integration of CAD tools [25, 28]. This development is as essential as achieving a paradigm shift in the actual integration technologies. An example of integrated analysis used to optimize the design of the active IPEM can be seen in the flowchart shown in Fig. 23. In this example, a software tool called iSIGHT is used to integrate and manage the data exchange between all other software tools. As can be seen, the user inputs the solid-body geometry and material data describing the IPEM layout into the mechanical CAD software, such as I-DEAS, which has the capability of exporting the geometry in a number of format files and an optional open architectural library that facilitates software integration. The same geometry and material data are shared by both the electromagnetic field and thermal analysis software packages.

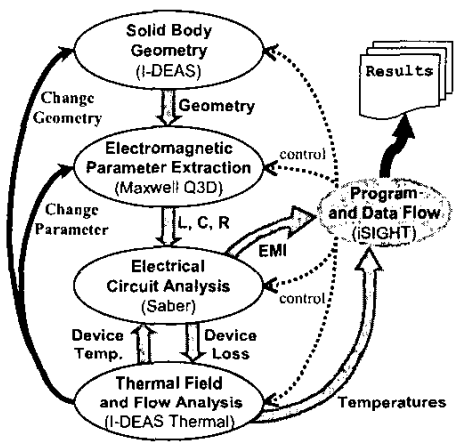


Fig. 23. Integrated thermal and electrical analyses.

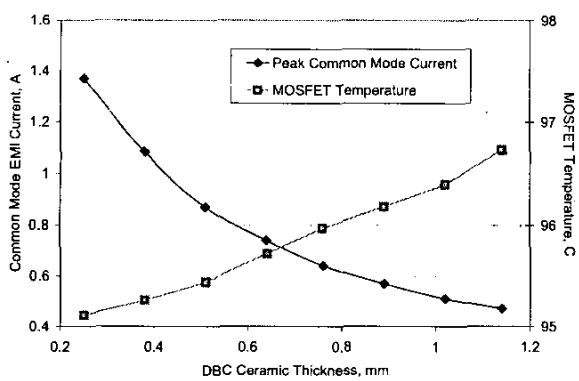


Fig. 24. Tradeoff between the common-mode EMI current and MOSFET junction temperature.

The structural parameters are extracted by Ansoft Maxwell and transferred as an equivalent circuit netlist to a circuit simulation software such as Saber that will then perform transient, EMI and loss analyses. I-DEAS carries out the thermal analysis to produce the temperature map within the IPEM. An interaction between Saber and I-DEAS is managed by iSIGHT until the temperature and device losses achieve convergence. This kind of analysis enables the designer to evaluate the tradeoffs between the electrical and thermal performance at the component, sub-system and system levels, such as the dependence of the device junction temperature and the peak common-mode EMI current on the thickness of the ceramic substrate, as shown in Fig. 24. The user can change the relative layout, size, and material of the structural parts, or even select different heat sink sizes or fluid flows, to achieve satisfactory results [28, 29].

In this example, Saber, Maxwell Q3D, and I-DEAS are separately controlled by batch files written in their own macro languages. Then, iSIGHT is used to achieve data exchange between these software tools by manipulating input and output files. The integrated system allows the simulations to be performed automatically, and 8 data points can be generated in less than 24 hours. The result shows that the common mode EMI current decreases and the MOSFET temperature increases as the DBC ceramic thickness increases, although the effect on the electrical performance is significantly greater than that for the thermal performance. The best selection of the DBC ceramic thickness was based on the trade-offs between electrical and thermal performance as well as the commercially available thickness.

VI. CONCLUSION

In the past, advances in semiconductor technologies have been the major driving force for reducing power electronic converter size, weight, and cost - mostly due to an increase in switching frequency. This increase in frequency and reduction in size have led to an increase in electromagnetic and thermal coupling in the layouts. Consequently an order of magnitude increase in switching frequency will require substantial reduction in structural inductances associated with device and system-level packaging, as well as require improved thermal characteristics. Therefore, to further improve performance, reliability, and reduce cost, it is essential to develop novel integration and packaging technologies in the form of Integrated Power Electronics Modules (IPEMs), which must enable the integration of all the converter functions and not only concentrate on the switching stage.

These novel integration and packaging technologies, in conjunction with suitable devices, sensors and integrated design tools used to exploit the physical properties of available materials have been the research focus of the Center for Power Electronics Systems. This paper has discussed some of these integration and packaging technologies and illustrated their advantages. The role of the technologies mentioned above in the IPEM concept is key to achieving high levels of integration and to enable significant growth of the power electronics industry. It is evident that further innovations will be necessary as power densities increase. The

impacts of system integration via IPEMs will enable a rapid growth of power electronics applications with reduced costs and design cycles that can be compared to the impacts in computer applications brought up by the VLSI circuit technology.

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